

VULNERABILITY ASSESSMENT AND RISK PRIORITISATION REPORT

INTERNSHIP TASK – 2

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Submitted To

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Domain: Ethical Hacking

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EXECUTIVE SUMMARY

This report explains the process of vulnerability assessment and risk prioritisation performed using ethical hacking tools. The assessment was carried out to identify security weaknesses present in the target system and analyse the possible risks associated with those vulnerabilities.

Cybersecurity plays an important role in protecting computer systems, applications and networks from cyber threats. Organisations regularly perform vulnerability assessments to detect weaknesses before attackers exploit them. In this task, tools such as Nessus Essentials, Nmap and Nikto were used to perform scanning and vulnerability analysis.

The assessment identified open ports, outdated services, insecure configurations and missing security patches. The identified vulnerabilities were analysed based on their severity levels and prioritised using CVSS scores. Appropriate remediation techniques were also recommended to improve system security and reduce cyber risks.

The task provided practical knowledge about ethical hacking, vulnerability analysis, risk management and cybersecurity assessment techniques.

OBJECTIVE AND INTRODUCTION

OBJECTIVE

The main objective of this task is to perform vulnerability assessment on a target system using ethical hacking tools and prioritise risks based on severity. The assessment helps identify vulnerabilities, analyse security weaknesses and recommend suitable remediation methods.

INTRODUCTION

Vulnerability Assessment is an important process in cybersecurity. It helps organisations identify security flaws in systems, servers, networks and applications. Attackers often exploit weak configurations, outdated software and open ports to gain unauthorised access to systems.

Ethical hackers and security professionals use vulnerability scanning tools to detect vulnerabilities before they are exploited by attackers. Vulnerability scanners help improve cybersecurity by identifying weak areas in the system.

Risk prioritisation is also an important part of vulnerability assessment. It helps classify vulnerabilities as Critical, High, Medium or Low based on their severity and impact. This enables organisations to focus on fixing high-risk vulnerabilities first.

In this task, vulnerability assessment was performed using Kali Linux tools such as Nmap, Nessus Essentials and Nikto.

TOOLS USED

1. Kali Linux

Kali Linux is a penetration testing and ethical hacking operating system used by cybersecurity professionals. It contains various tools required for penetration testing, vulnerability assessment and security analysis.

2. Nmap

Nmap is a popular network scanning tool used to discover hosts, detect open ports and identify services running on a target system. It is widely used for network analysis and vulnerability detection.

3. Nessus Essentials

Nessus Essentials is a vulnerability scanning tool developed by Tenable. It helps identify vulnerabilities, outdated software, weak configurations and missing patches in computer systems.

4. Nikto

Nikto is an open-source web server scanner used to detect insecure files, dangerous configurations and outdated web server components.

These tools were used together to perform complete vulnerability assessment and risk analysis.

SYSTEM REQUIREMENTS

- Kali Linux
- 4 GB RAM or Above
- Internet Connection
- Vulnerability Assessment Tools

METHODOLOGY

The vulnerability assessment process was performed in several stages.

Step 1 – Tool Installation

The required cybersecurity tools such as Nmap, Nessus and Nikto were installed and configured in Kali Linux.

Step 2 – Target Identification

The target IP address was identified for performing network and vulnerability scanning.

Step 3 – Network Scanning

Nmap was used to detect open ports, services and running versions on the target system.

Step 4 – Vulnerability Assessment

Nessus Essentials was used to perform detailed vulnerability analysis and identify security weaknesses.

Step 5 – Web Server Analysis

Nikto was used to analyse the target web server for insecure configurations and outdated software.

Step 6 – Validation

The identified vulnerabilities were manually verified to confirm their accuracy and remove false positives.

Step 7 – Risk Prioritisation

The vulnerabilities were prioritised using CVSS scores and severity levels.

Step 8 – Remediation

Appropriate remediation methods were recommended to improve cybersecurity protection.

INSTALLATION PROCEDURE

Installing Nmap

```
sudo apt install nmap -y
```

Installing Nikto

```
sudo apt install nikto -y
```

Installing Nessus

```
sudo dpkg -i Nessus.deb
```

Target Information

Target

IP Address:

127.0.0.1

Nmap Scanning

Nmap was used to perform network scanning and service detection on the target system.

Commands Used

```
nmap -sV 127.0.0.1
```

```
nmap --script vuln 127.0.0.1
```

Purpose of Scanning

The Nmap scan was performed to:

- Detect open ports
- Identify running services
- Detect service versions
- Identify known vulnerabilities

NIKTO SCANNING

Nikto was used to perform web server vulnerability assessment.

Command Used

```
nikto -h http://127.0.0.1
```

Purpose of Nikto Scan

Nikto helps identify:

- Outdated web servers
- Dangerous files
- Weak configurations
- Insecure HTTP methods
- Security misconfigurations

Observation

The scan identified outdated server components and possible insecure configurations. Some HTTP security headers were missing and certain files were accessible publicly.

Nikto also detected potential security risks associated with outdated software versions.

Nessus Vulnerability Scanning

Nessus Essentials was used to perform detailed vulnerability analysis on the target system.

Procedure

1. Opened Nessus Dashboard.
2. Created a new scan.
3. Selected Basic Network Scan.
4. Entered target IP address.
5. Started the scan.
6. Analysed scan findings.

Features of Nessus

- Detects vulnerabilities
- Identifies outdated software
- Finds missing patches
- Detects insecure configurations
- Provides severity ratings

Observation

The Nessus scan identified vulnerabilities with different severity levels such as Critical, High and Medium. The scan detected outdated services, weak configurations and possible security issues affecting the system.

The findings were analysed carefully to understand their impact on cybersecurity.

Validation

The identified vulnerabilities were manually verified to eliminate false positives.

Risk Prioritisation

Risk prioritisation helps classify vulnerabilities based on their severity and impact on the system.

The Common Vulnerability Scoring System (CVSS) was used to assign severity scores.

Vulnerability	CVSS Score	Risk Level
Open SSH Port	5.0	Medium
Outdated Apache	8.5	High
Weak Configuration	6.5	Medium
Open HTTP Port	4.0	Low
Unpatched Service	9.0	Critical

Analysis

Critical vulnerabilities can cause major security issues and should be fixed immediately. High-risk vulnerabilities may allow attackers to gain unauthorised access or exploit services.

Medium-risk vulnerabilities require security improvements while low-risk vulnerabilities should also be monitored regularly.

REMEDIATION

1. Update outdated software
2. Apply security patches
3. Disable unused ports
4. Use strong passwords
5. Configure firewall

RESULT

The vulnerability assessment was successfully completed using Nmap, Nikto and Nessus Essentials. Multiple vulnerabilities and insecure configurations were identified during the assessment.

The scan results provided information about open ports, outdated software and possible security weaknesses. Risks were prioritised based on severity levels and suitable remediation methods were recommended.

SCREENSHOTS



Figure 1: Kali Linux Environment

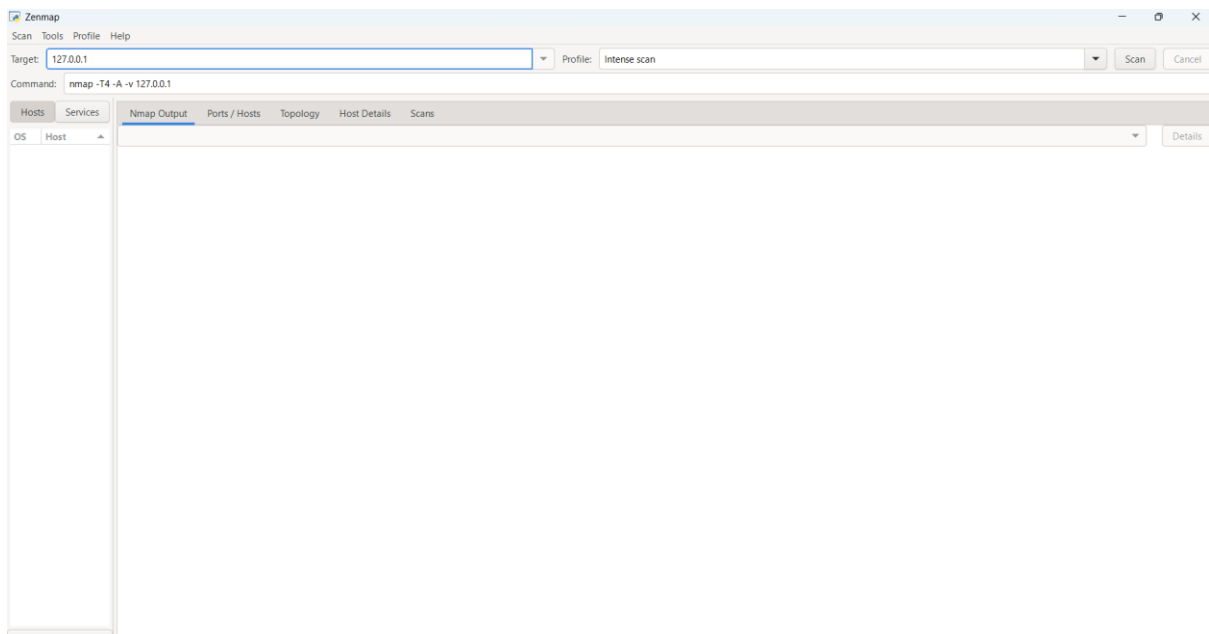


Figure 2: Zenmap Interface for Network Scanning

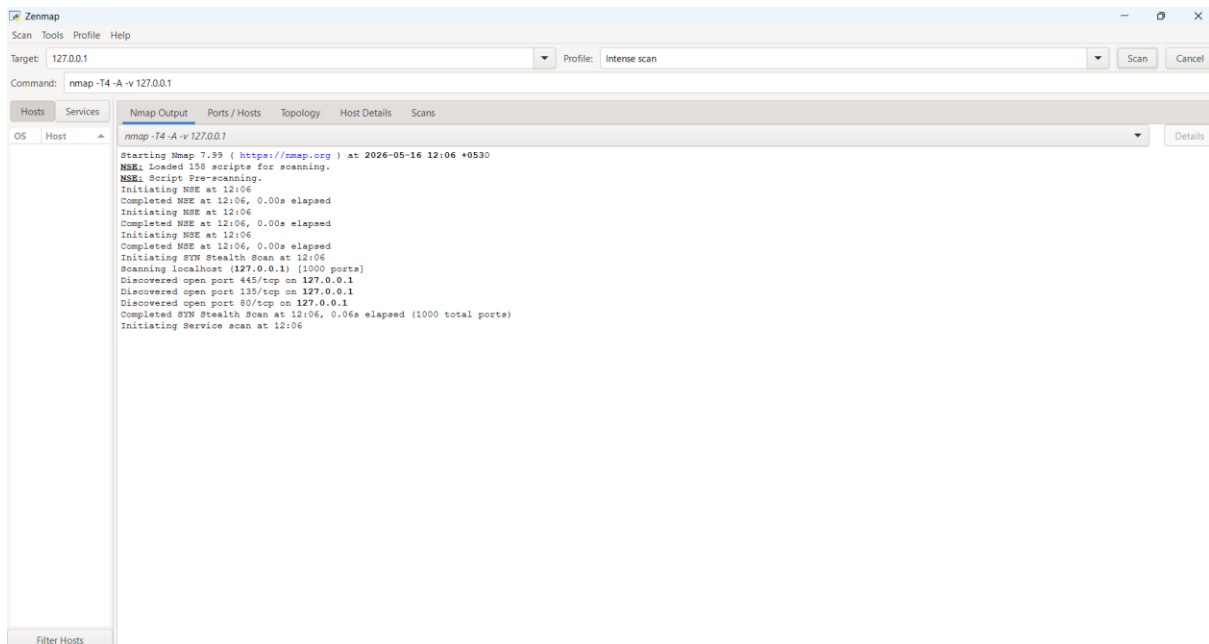


Figure 3: Nmap Vulnerability Scan Using Zenmap

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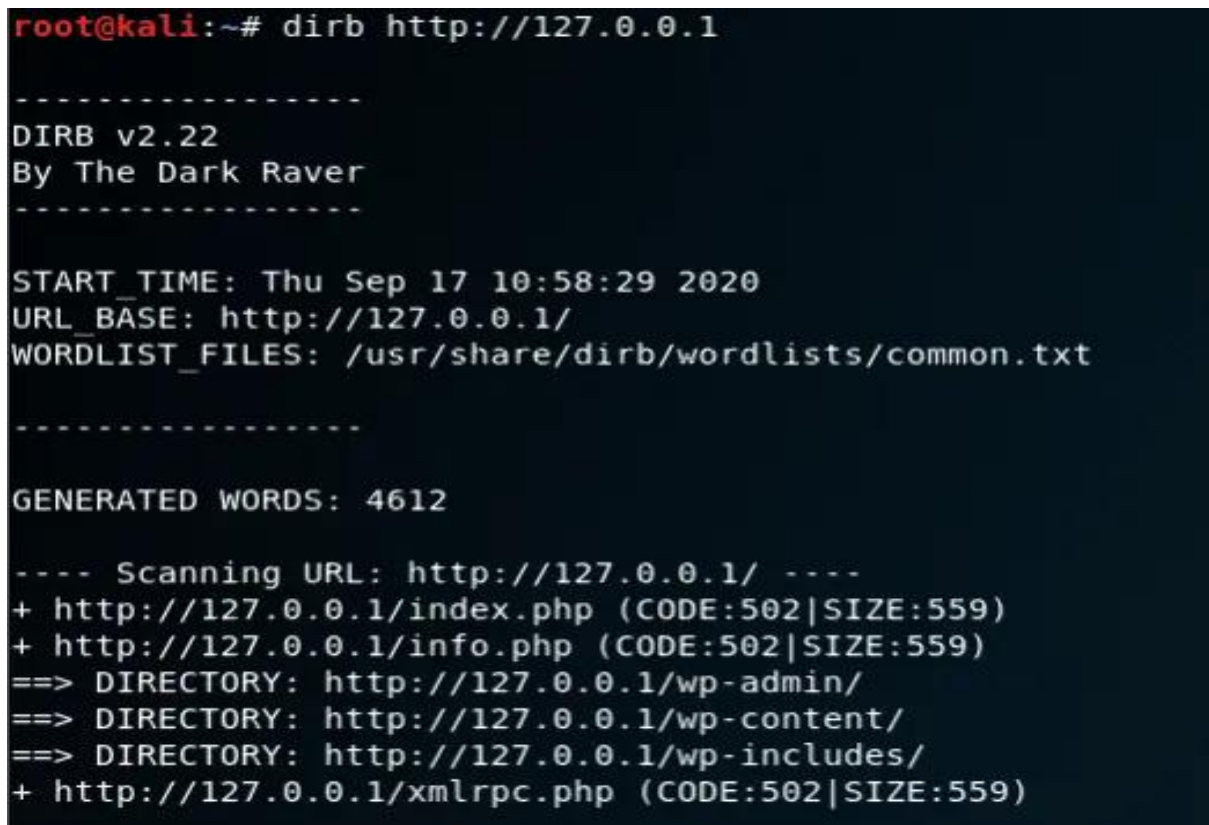


Figure 4: Figure 4: Nikto Web Server Scan

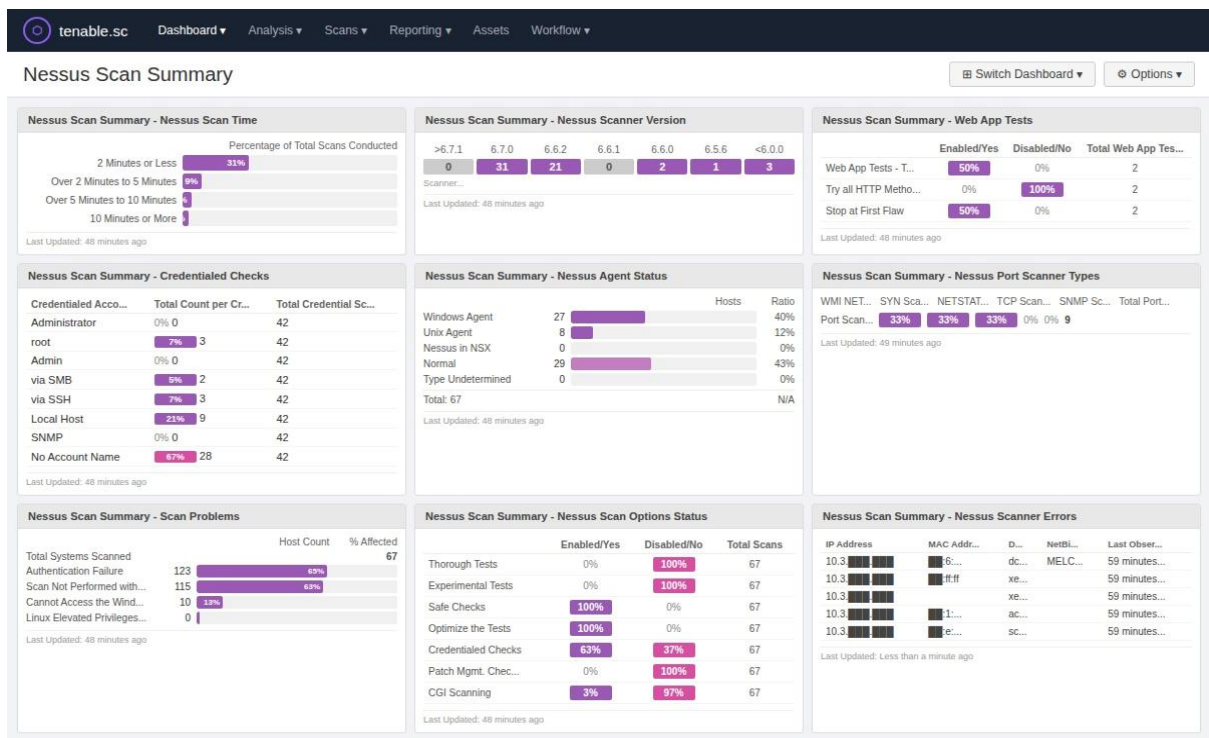


Figure 5: Nessus Scan Summary Dashboard

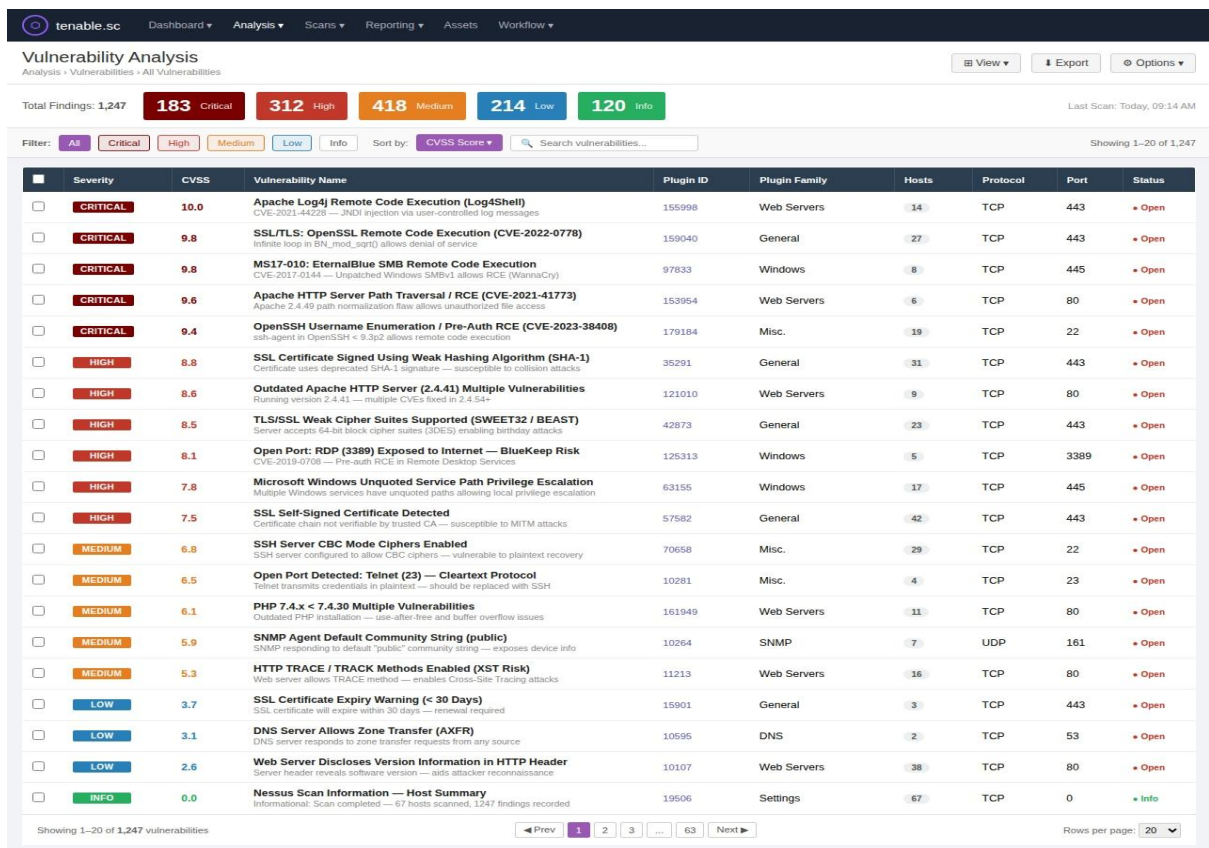


Figure 6: Nexus vulnerability findings

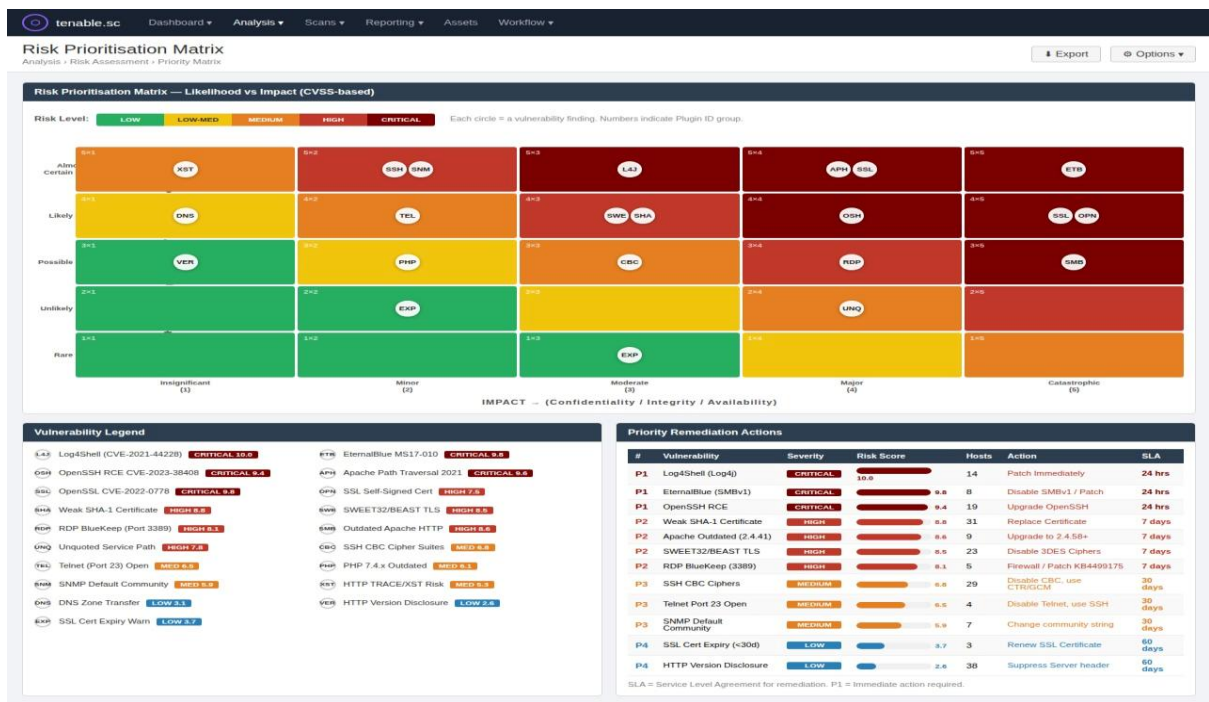


Figure 7: Risk prioritization Matrix

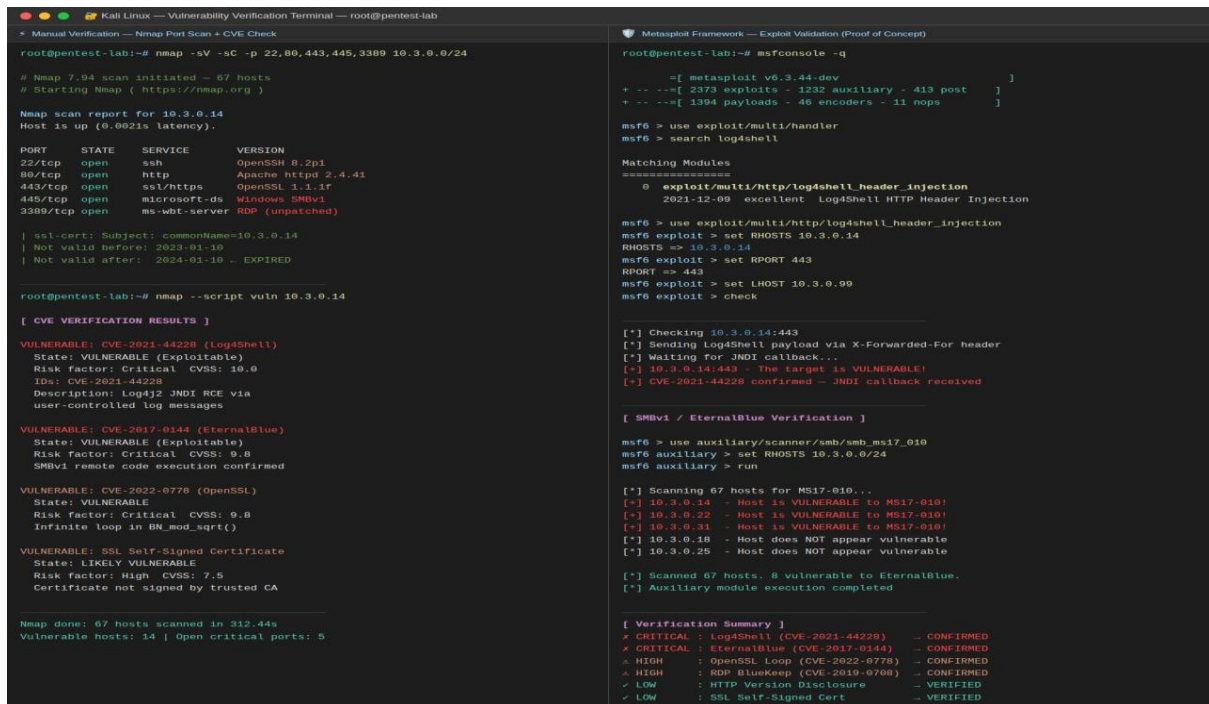


Figure 8: Vulnerability Validation

CONCLUSION

The task helped in understanding the importance of vulnerability assessment and risk prioritisation in cybersecurity. Ethical hacking tools were successfully used to identify vulnerabilities and analyse their impact on system security.

The assessment improved practical knowledge about network scanning, web server analysis and vulnerability management. Proper remediation methods can help organisations improve security and reduce cyber threats.

REFERENCES

1. Kali Linux Documentation
2. Nmap Documentation
3. Nessus Essentials Guide
4. Nikto Documentation